Edward Y. Chang
Department of Electrical & Computer Engineering
Department of Computer Science (affiliated)
University of California
Santa Barbara, California 93106
Phone: (805) 893-2917
Email: echang@ece.ucsb.edu

EDUCATION

Ph.D.	Stanford University, Electrical Engineering (Advisor: Hector Garcia-Molina)	1995-99
M.S.	Stanford University, Computer Science	1992-94

AWARDS

NSF Career Award	2002
IBM Faculty Partnership Award	2000, 2001, 2002

EXPERIENCE

Associate Professor (with tenure), University of California, Santa Barbara	2003-present
Co-Founder & CTO, VIMA Technologies, Santa Barbara	2001-present
Assistant Professor, University of California, Santa Barbara	1999-2003
Staff Engineer, Sun Microsystems, Mountain View, CA	1994-95

SELECTED PATENTS

"*Kernel Alignment for Learning Imbalance Datasets*," with University of California., 2003.
"*Virtual IO: Preemptible Disk Access*," with University of California., 2002.
"*DPF: Dynamic Partial Function*," with VIMA Technologies Inc., 2002.
"*Perception-Based and Personalizable Image Retrieval*," with University of California, 2001.
"*Personalizable Interactive Digital TV*," with University of California, 2001.

PROFESSIONAL ACTIVITIES

Associate Editor, for IEEE Transactions on Knowledge and Data Engineering 2003-
Guest Editor, for ACM Multimedia System Journal Special Issue on Video Surveillance 2004
Co-Chair, for ACM International Workshop on Video Surveillance 2003/2004
Chair, for IEEE Pacific-rim Conference on Multimedia Special Session on Video Surveillance 2003
Associated PC Chair, for ACM MM 2002/2003/2004
Panelist, for National Science Foundtion 2002/2003
PC Member, for ACM SIGMOD 2004, SIAM Data Mining 2004, ACM CIKM 2003, IEEE MM 2002/2003/2004, WISE 2002/2003, ICCV 2003, ICDE 2002, PADKDD 2002, etc.
Local Arrangement Chair, for ACM SIGMOD 2001.

SELECTED INVITED TALKS

- "*Sequence-Data Learning for Video Surveillance*," University of Singapore, December 2003.
- "*SFINX: A Multi-sensor Fusion and Mining System*," IBM Almaden & T.J. Watson Centers, June, September 2003.
- "*Multimedia Semantics and Machine Learning*," Tutorial, International Conf. on Multimedia, Baltimore, July 2003.
- "*Statistical Learning Methods for Emerging Database Applications*," Plenary Tutorial, International Conf. on Database Systems for Advanced Applications, Kyoto, March 2003.
- "*Multimedia Information Retrieval*" Panel, International Conf. on Multinedia, Sept. 2002.
- "*Multimedia Data Mining*," HP Labs., December 2001; January 2002.
- "*Dynamic Partial Function*," UC Berkeley Digital Library Seminar, February 2002.
- "*Measuring and Learning Perceptual Similarity*," UCLA, October 2001.
- "*Measuring and Learning Perceptual Similarity*," INRIA-Rocquencourt, September 2001.

TEN RECENT PUBLICATIONS (ALL RELATED)

- *"Multi-camera Spatio-temporal Fusion and Biased Sequence-data Learning for Security Surveillance,"* G. Wu, Y. Wu, L. Jiao, Y.F. Wang, and E. Chang, ACM International Conference on Multimedia, pp. 528-538, Berkeley, November 2003 (17% accepted).
- *"Confidence-based Dynamic Ensemble for Image Annotation and Semantics Discovery,"* (best student paper finalists), B. Li, K.-S. Goh, and E. Chang, ACM International Conference on Multimedia, pp. 195-206, Berkeley, November 2003 (17% accepted).
- *"Adaptive Feature-Space Conformal Transformation for Imbalanced-Data Learning,"* G. Wu and E. Chang, Internation Conf. on Machine Learning (ICML), Washington DC, Aug. 2003.
- *"CBSA: Content-based Soft Annotation for Multimodal Image Retrieval Using Bayes Point Machines,"* E. Chang, K. Goh, G. Sychay, and G. Wu, IEEE Transactions on Circuits and Systems for Video Technology Special Issue on Conceptual and Dynamical Aspects of Multimedia Content Description, 12(1), pp.26-38, January 2003 (16% accepted).
- *"MEGA: The Maximizing Expected Generalization Algorithm for Learning Complex Query Concepts,"* E. Chang and T. Li, in ACM Trans. on Information Systems, 2003.
- *"The Discovery of A Percpeptual Distance Function for Measuring Image Similarity,"* B. Li, E. Chang, and Y. Wu, ACM Multimedia Journal Special Issue on Content-Based Image Retrieval, Volume 8, Number 6, pp. 512-522, April 2003 (12% accepted).
- *"DynDex: A Dynamic and Non-metric Space Indexer,"* K. Goh, B. Li, and E. Chang, in Proc. of the 9^{th} ACM International Conf. on Multimedia, pp.466-475, December 2002 (14% accepted).
- *"Approximate Similarity Queries in High-Dimensional Spaces,"* C. Li, E. Chang, H. Garcia-Molina, and G. Wiederhold, IEEE Transactions on Knowledge and Data Engineering, Volume 14, Number 4, pp.792-808, July/August 2002.
- *"Answering Queries with Useful Bindings,"* C. Li and E. Chang, ACM Transaction on Databases, Volume 2, Number 3, pp.313-343, September 2001.
- *"Support Vector Machine Active Learning For Image Retrieval,"* S. Tong and E. Chang, in Proc. of the 9^{th} ACM International Conf. on Multimedia, pp.107-118, Ottawa, October 2001 (16% accepted).

INVITED PAPERS

- *"Statistical Learning for Effective Visual Image Retrieval,"* E. Chang, B. Li, and G. Wu, IEEE International Conf. on Image Processing, Barcelona, September 2003.
- *"On Learning Perceptual Distance Functions for Image Retrieval,"* B. Li and E. Chang, in Proc. of IEEE International Conf. on Acoustics, Speech and Signal Processing, Orlando, May 2002.
- *"High-dimensional Visual Data Indexing,"* K. Goh and E. Chang, to appear in Proc. of the 6^{th} International Conf. on Visual Databases, Australia, May 2002.
- *"Learning and Measuring Perceptual Similarity,"* E. Chang, in Proc. of the 1^{st} NSF/INRIA/Berkeley/IBM Workshop on Content-based Multimedia Information Retrieval, Paris, September, 2001.

COLLABORATORS & GRADUATE STUDENTS

Graduate Advisor: Hector Garci-Molina; *Collaborators*: Chen Li, Simon Tong, James Wang, Yuan-Fang Wang, Gio Wielderhold; *Graduate Advissees*: Zoran Dimitrijevic, Kingshy Goh, Ankur Jain, Beitao Li, Yan Meng, Arun Qamra, Navneet Panda, Raju Rangaswami, Gerard Sychay, Gang Wu, Yi Wu, Huaxin You.